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Cited Text Span Extraction for Scientific Question Answering

Bachelor's Thesis

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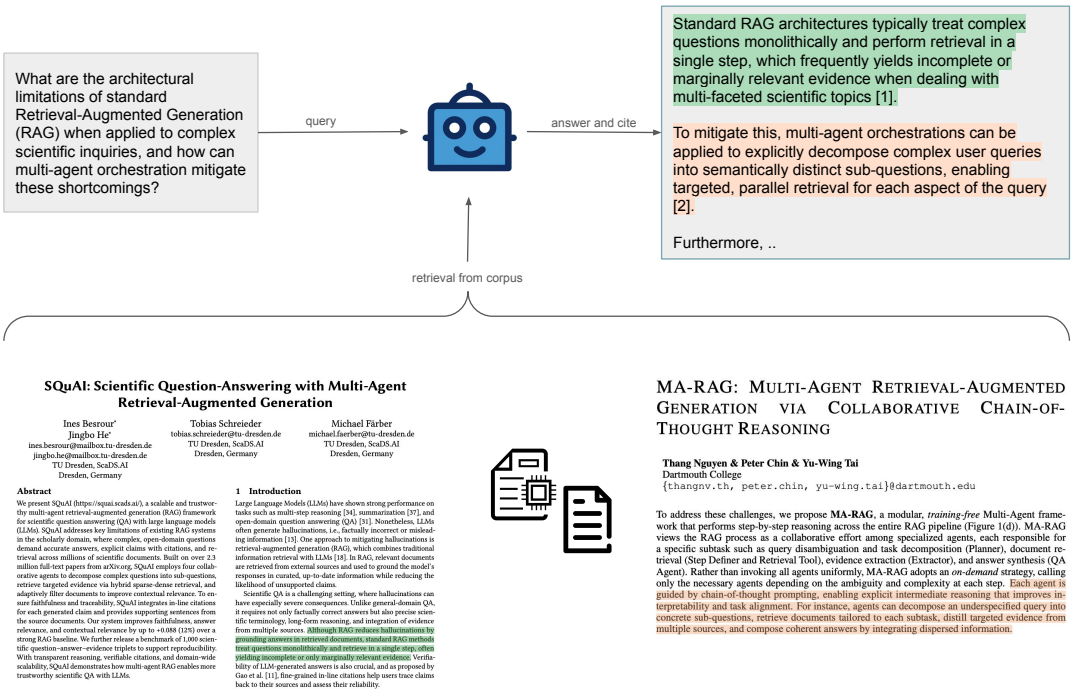
1. Introduction

Large Language Models (LLMs) have fundamentally transformed how we look up information and obtain answers to questions, becoming an increasingly important tool, not only in everyday life, but also across a range of professional fields. With the usage of such systems, the need to verify the factual correctness and easily do further research on the underlying sources arises, especially when answers lack important details or provide a depth of explanation that is misaligned with the user's existing knowledge. This is especially important in high-stakes fields such as medicine, law or in scientific research, where seemingly plausible but unbacked or false claims could lead users to draw false conclusions.

One increasingly popular method is the approach of Retrieval Augmented Generation (RAG), replacing zero-shot LLM solutions with a two-step process. Firstly, relevant sources in the form of documents or other multi-media data are retrieved and then fed into the generating model for it to use as ground truth for answer generation and later to enable citation extraction. Retrieving correct and precise text-spans alongside the generated answer is not a trivial task and with domain-specific requirements widely varying such systems are part of active research [1] [2]. One system applying such a RAG approach is SQuAI [3], aimed at answering complex open domain scientific questions on its knowledgebase containing 2.3 million full-text scientific papers. For further performance improvements it introduces a multi-agent pipeline, enhancing document retrieval capabilities as well as providing document level backlinks for parts of its generated answer. This system, while providing some backlinking, does still fall short of providing more granular, sentence-level cited text spans, which have proven to be much more useful to users, as they are quicker and easier to verify, enhancing trustworthiness and usability of the system.



Therefore this thesis will focus on benchmarking different cited text span extraction methods to enhance source attribution, leading to the following core contributions:

1. Curate a dataset of high quality, multi-hop scientific questions, building upon a large-scale retrieval corpus, that require complex reasoning capabilities in order to be answered and closely resemble real world scientific inquiry needs.

2. Introduce a hierarchy of different cited text span extraction methods ranging from simple lexical and semantical matching up to complex reasoning capabilities, as well as strategic combinations of such, to allow for tradeoffs in accuracy and efficiency.

3. Apply an automated framework to benchmark the retrieval and answering capabilities of non-parametric, retrieval- and evidence-based text generation systems on the proposed dataset as well as evaluating the extraction methods in terms of efficiency and correctness and quality of source attribution using NLI models and LLM-as-a-judge approaches.

2. Related Work

2.1. Datasets

In order to improve upon the existing SQuAI system, a highly specialized dataset of open domain, long form scientific questions is needed. Due to the scarcity and high cost of curating such datasets, only a limited number of suitable options are available. With SQuAI being designed as a system to answer complex scientific inquiries, based on the UnArXive2024 Dataset [4], further requirements are introduced, namely:

1. Domain Alignment: Questions must fall under the knowledge scope of the underlying knowledgebase, in order for the system to be able to provide cited text spans as proof alongside its answer. In terms of the SQuAI System this boils down to questions targeting physics, mathematics and computer science that are also part of the papers in the UnarXive 2024 Dataset.

2. High Complexity: Questions must go beyond simple factoid questions and require complex reasoning capabilities to answer, as questions posed by scientists usually require synthesizing, interpreting and evaluating data, requiring high cognitive abilities to interact with. In order to be able to test if the SQuAI system can be used under such circumstances, the questions in the dataset must also have a similar degree of complexity.

3. Multi-Hop: Questions must require the information in multiple documents to be answered, in order to test the ability of the system to synthesize information from multiple sources, as is required in naturally occurring scientific research.

4. Decontextualization: Questions must be answerable in an open domain settings and cannot contain any kind of reference to specific papers and their contents that lose meaning or become ambiguous outside of the context of their paper. As a result questions must contain enough context such that they are standalone answerable.

5. Text format: Questions can only target information in a paper that is present in the form of text, as current versions of the SQuAI system do solely rely on textual information, not taking into account images, graphs or similar.

6. Ground Truth: In order to evaluate the correctness of the model's answer, a dataset must provide a ground answer alongside each question.

However, comparing some of the most prominent datasets shows that none fits all discussed criteria, finally resulting in the need for own curation.

Dataset	Domain Alignment	High Complexity	Multi-Hop	Decontextualization	Text format	Ground Truth	Sufficient Dataset size (>500)
Qasper [5]	partial	shallow	no	no	yes	yes	yes
ALCE [6]	no	shallow	yes	yes	yes	yes	yes
SPIQA [7]	yes	complex	no	partially	no	yes	yes
QASA [8]	yes	complex	no	no	yes	yes	yes
ExpertQA [9]	yes	very complex	yes	yes	yes	yes	no [1]
SciDQA [10]	yes	very complex	no	no	no	yes	yes
PeerQA [11]	yes	very complex	no	no	yes	yes.	yes

Table 2.1.: Comparison of Scientific QA Datasets against SQuAI Requirements
Abbreviations: CS = Computer Science; ML = Machine Learning; NLP = Natural Language Processing.

2.2. Cited Text Span Extraction

Context extraction methods can be most broadly divided into pre- and post-hoc extraction methods, depending on the time the context attribution happens. While pre-hoc systems such as QASPER [5] and QASA [8] extract citable passages before and during runtime of the question-answering system, post-hoc systems search and attach supporting evidence afterwards. Post-hoc systems have been shown to outperform pre-hoc systems, achieving higher citation coverage and answer correctness as well as lowering model hallucination [12]. While a number of systems rely on zero shot LLM solutions for context extraction [13] [14] [15], work on the DECOMPTUNE [16] system has shown that decomposing facts and decontextualizing answer sentences before attributing source text spans with LLM prompting, can yield significant improvements, especially when applied to answers for long-form, multi-hop questions. In order to eliminate the probability of hallucinated model output, models can be finetuned [17] or answer space restricted to the start and end token of the cited text span, forcing strict citation [18].

So called white-box methods rely on access to model's internal weights, rendering them incompatible with proprietary LLMs. The LAQuer [19] system as well as earlier token-embedding approaches it is built upon [20], analyzes internal hidden states to extract the most likely text span, a user-highlighted part of the LLM's answer is based upon. To pinpoint a certain text span, the system calculates the maximum cosine similarity between individual tokens in the source text and the averaged hidden representation of the highlighted span, for which attribution is sought. The system then extracts the precise quote from the surrounding window of text. However, this method has been shown by the authors to perform significantly worse than LLM Prompting, with drastically lower accuracy and missing suitable citations, especially in larger search spaces e.g. longer text. Similarly AttnAttribute [21] identifies "Attention Heads", tracking the model's internal attention values during generation time. Just like LAQuer it then extracts text spans from the passage surrounding the anchor with the highest model attention. Benchmarks made as part of the development of DECOMPTUNE [16] concluded that both of those methods performed significantly worse than LLM Prompting.

Recent advancements have explored generative approaches. Rather than retrieving and reranking text spans, such systems generate them. In order to prevent hallucinations, the output vocabulary

[1] question answer pairs are broadly distributed across diverse fields of stuy, resulting in insufficient topical overlap with unArXive2024

of the systems is typically restricted to the underlying retrieval corpus by applying constrained decoding mechanisms, such as prefix trees or FM-indices [22]. GERE [23], as one such a system using a prefix tree, applies this concept of restricted output in the document retrieval step, generating document titles for selecting appropriate source texts, as well as sentence identifiers, mapped to individual sentences. This system, while removing the need for a massive document index as well as showing significant improvements in precise and correct text span extraction compared to baselines such as Dense Passage Retrieval (DPR) and BM25, requires the prefix tree to be precomputed and stored in memory, making the systems highly inflexible for changes in the underlying corpus. Similarly MRR-FV [24] further improves this approach with multi-hop reasoning capabilities. Instead of generating sentence identifiers, it generates sentences directly, leveraging the natural language understanding capabilities of LLMs, leading to improvements in precision and quality of attributed spans, but also further bloating the prefix tree, making the system even harder to scale.

Despite the diversity of proposed paradigms, current work mostly evaluates them in isolation, restricting comparison to baseline or direct predecessors on heterogeneous tasks, making broad comparisons for practical applicability unfeasible. Consequently, this work introduces a hierarchy of extraction methods, increasing in both reasoning capabilities and computational requirements, benchmarking both accuracy and efficiency on a unified framework, allowing for systematic trade-off analysis between paradigms in the context of open-domain multi-hop scientific QA.

2.3. Span Evaluation

While human annotation has always been the gold standard for assessing natural language generation (NLG) tasks, relying solely on it is a bottleneck for developing models of increasing size and complexity. While providing feedback with extensive reasoning and natural language understanding, human evaluation suffers from being costly, time intensive and, depending on the specific task, highly subjective. Thus lowering or even replacing the need for human evaluation completely is an important problem.

Early metrics heavily rely on measuring the lexical overlap between text spans, for instance of a human-written ground truth and an LLM generated answer for the same questions. The most prominent examples include BLEU [25] and ROUGE [26], both being n-gram based metrics. While the former is precision-oriented, measuring how many of its n-grams also appear in the human reference span, the latter applies a recall-oriented approach, measuring how many n-grams of the human reference are also present in the LLM generated text span. Being computationally inexpensive and language independent, these metrics have established themselves as rigid industry standards and baselines for other methods. Their alignment with human judgement, however, has been shown to be low, suffering from the lack of semantic understanding, leading to penalizing valid paraphrasing as well as missing contradictory or hallucinated content.

Partly overcoming those limitations, embedding-based metrics like BERTScore [27] and MoverScore [28] emerged, measuring semantic instead of lexical similarity. BERTScore does one-to-one greedy matching on vector embeddings, providing precision, recall and F1 score metrics, just like BLEU and ROUGE, but on an embedding level. MoverScore improves upon this idea, utilizing Earth

Mover's Distance (EMD) allows for introducing one-to-many relationships, allowing for matching of single words with phrases, so called soft matching, as well as introducing Inverse Document Frequency (IDF) for enhanced lexical matching, to better reflect the importance of specific keywords to be included. Though significantly more expensive to compute, those metrics show significant improvements in semantic robustness as well as intrinsic context awareness, due to their vector embeddings. However, both of those metrics still require a reference answer to compare to. BartScore [29] eliminates this requirement, measuring the conditional probability one text span is generated from another using an encoder-decoder model, arguing that the probability is higher, if the claims in the answer are also deeply rooted in the source text.

Besides similarity-based metrics, other research suggests, to approach the task from a logical entailment perspective. The AIS [30] framework proposed a unified conceptual definition for such, arguing that a statement is considered Attributable to Identified Sources (AIS) iff completely corroborated by a source. In order to verify this the authors imposed additional constraints being 1) that the claim made in an answer sentence must be standalone interpretable, thus decontextualized and 2) the information in the text span must support all the information present in the answer span it is attributed to. Previously discussed similarity based metrics like BLEU, ROUGE or token-level F1, as well as solely relying on an out-of-the-box Natural Language Inference (NLI) model have been shown to have low correlation with human judgement for both factual consistency and logical entailment assessment [31] [32]. To address these limitations systems relying on NLI models like FactCC [32] argued that evaluating logical entailment on sentence-to-sentence level is insufficient, since information in sources might be spread across multiple passages. As a result these systems increased their evaluation scope to whole source text, leading to better judgement capabilities, but also increase in model input. The authors of FactCC observed that a majority of misclassifications were commonsense mistakes; the TRUE benchmark [31] observed improvements with increasing model size but also degradation with longer input, which is further supported by research on the SummaC [33] framework. To shrink input size, SummaC increased evaluation granularity by dividing both source text and model answer into smaller units, checking each combination of them for entailment and combining the obtained scores.

The Q^2 system [34] proposed using a Question-Generation - Question-Answering (QG-QA) method, generating questions to which the LLM generated answer is the solution. Those questions are then answered with the original source document and lastly the so obtained answers are then compared with the original LLM answer using a NLI model. The authors presented significant improvements compared to baseline methods, further enhanced by the TRUE benchmark by applying an ensemble approach utilizing ANLI (standard NLI), SummaC (increased granularity NLI) as well as the Q^2 (QG-QA) method. The ensemble was shown to beat every standalone evaluation metric previously discussed on all datasets benched in TRUE, emphasizing the need for combined, complementary evaluation metrics.

Recently the evaluation paradigm has shifted to LLM-as-a-judge approaches. While early versions like GPTScore [35] tried evaluating based on model internals, using the conditional token probability, thus the confidence of a model in a given answer, they have quickly been shown to be much less effective than prompting the LLM directly to grade [36]. The performance of LLMs in

evaluation tasks can additionally be enhanced by applying a Chain-of-thought approach, prompting the model to break down complex tasks into simpler ones and adding intermediate reasoning steps. Closely resembling human cognitive processes, this method yields significant improvements, especially with larger models and problems of increasing complexity [37], reaching agreement rates comparable to human judges for open-ended multi-turn dialogues [38], thus eliminating human annotation constraints.

When applied to long-form text containing multiple pieces of information, binary judgements become insufficient as they cannot reflect partial attribution. FactScore [39] addresses this by breaking down answers into atomic facts using an LLM, obtaining a percentage measure of supportedness. AttrScore [40] introduces additional evaluation criteria, such as measures for attribution, extrapolation and contradiction, in order to further characterize the type of attribution error. Similarly, CiteEval [41] introduces penalties for attributions containing unnecessary noise and redundant citations while advocating for evaluation against full source documents, in order to not lose information in the surrounding context. The RAGAS [42] framework operationalized these findings, providing a suite of automated metrics like faithfulness, answer relevance and context relevance, without requiring human annotation. However, the importance of penalizing undesired behaviors is further emphasized by the MT-Benchmark [38], showing that LLMs have a multitude of biases, such as favoring longer and LLM generated text spans or switching preferences between answers when presented in a different order. Furthermore, LLMs perform remarkably poorly with fine-grained information, with a majority of mistakes being caused by overlooking numbers and entities or hallucinating logical connections [43]. Using increasingly large models additionally introduces high computational cost requirements. To address this, systems like ARES [44] rely on LLM generated, domain specific, synthetic training data to finetune smaller models. Using Prediction-Powered-Inference (PPI) ARES leverages a small human annotated dataset, to calculate the LLM's annotation error on the synthetic data, the "rectifier", to provide statistical confidence bounds and enhance its evaluation, beating existing zero shot solution like RAGAS.

Automatic evaluation of cited text span attribution remains a challenging task. Despite the wide range of approaches, no single metric being capable of reliably and comprehensively assessing attribution performance as a whole has emerged. Increasingly complex tasks, such as multi-hop questions and challenging application domains like scientific QA introduce further evaluation dimensions. Consequently, combining multiple evaluation metrics with complementary strengths becomes necessary to measure task-specific aspects within a unified framework.

3. Methodology and Approach

3.1. Dataset

To curate a dataset of high-quality, expert-level scientific question-answer pairs, a fully automated pipeline is applied. It consists of three consecutive steps: document selection, question answer generation and document usage verification, in order to ensure that generated questions and answers are scientifically rigorous and strictly dependent on multi-document information synthesis to be answered, as well as reliably providing associated ground truth documents alongside.

3.1.1. Document selection

To match the underlying document corpus, selectable papers are restricted to those of the unArXive2024 dataset, yet in theory this could be expanded at will. A randomly selected anchor paper as well as its cited references serve as the foundation for generating a single question-answer pair, ensuring broad coverage across the whole dataset. For each paper, a document-level SPECTER [45] vector embedding is retrieved via the Semantic Scholar academic graph API [46]. Leveraging citation graphs, these vectors provide dense representations of a paper's title and abstract, which are subsequently used to calculate the cosine similarity and thus the semantic similarity of each paper with the anchor document. To ensure topical alignment while excluding highly redundant work, a subset of candidate papers with a cosine similarity between 0.7 and 0.9 is selected. Four of these papers are then chosen at random to be used alongside the anchor paper for further generation. Restricting the model input to five papers in total provides a multitude of topical overlaps for the model to choose from without resulting in exceedingly long prompts, mitigating the risk of "Lost in the Middle" effects. If a chosen anchor paper fails to provide at least four appropriate references, it is considered insufficient to be used in the dataset.

3.1.2. Question-Answer generation

While modern LLMs offer increasingly large context windows, usually even above 100,000 tokens, empirical studies have shown severe performance degredation in tasks like complex reasoning, data retrieval and logic problems, accompanied by drastically increased hallucination [47]. In the generation step, the pipeline applies a Qwen/Qwen2.5-72B-Instruct [48] model, shown to have an effective context size with minimal degredation of about 32k tokens [49]. Although the model can be changed at will, for this specific configuration, the input should not exceed 128,000 [2] characters, thus 25,000 characters per paper - the base limit. With papers being highly variable in length, an adaptive per paper character limit is applied, distributing leftover characters from papers shorter than the base

[2] in standard heuristics for scientific english one token translates to roughly four characters

limit, weighted by paper length. Thus, the adaptive character limit for papers with a length greater then the base limit can be determined by:

$$C_{adaptive} = \left\lfloor C_{base} + \left(\frac{l_{paper}}{\sum_{p \in P_{above}} len(p)} \right) \times \sum_{s \in P_{below}} (C_{base} - len(s)) \right\rfloor \tag{3.1}$$

Paper texts longer than their adaptive character limit are then head and tail truncated, retaining the first and last $\frac{C_{adaptive}}{2}$ characters which are usually the most information dense for scientific papers.

Subsequently the LLM is prompted using a two-step strategy. Firstly the model is tasked with extracting topic overlaps between papers, identifying important concepts and entities as well topical relationships between documents. The following step is the generation step, where the model is prompted to generate one high quality and complex question-answer pair targeting a subset of the previously extracted concepts or entities. The model is incentivized to avoid simple definitional questions, instead opting for complex inquiries, that require information synthesis and evaluation across multiple source documents in order to be answered. Additionally, the model is instructed to answer the question as precisely as possible, providing a precision-biased subset of the underlying papers used during generation alongside. The complete prompt instructions are provided in Listing A.1.

3.1.3. Document usage verification

Generative LLMs have been shown to be notoriously prone to hallucinating citations and falsely attributing sources. Thus, their ability to reliably track a paper's actual contribution to the generated question-answer pair is drastically reduced. In order to mitigate this and ensure the dataset's integrity, a verification step is applied, employing a secondary LLM, specifically mistralai/Mixtral-8x7B-Instruct-v0.1 [50], to act as a judge for the actual usage of each paper individually. Utilizing a Chain-of-Thought (CoT) approach, the model is first instructed to identify core scientific claims and topics in the question-answer pair, before determining if a paper provides explicit evidence or data beyond mere keyword overlap, supporting the claims made in the answer, resulting in a final binary verdict for each paper. The complete prompt instructions are provided in Listing A.2.

The full dataset is then obtained by applying a consensus filter between both generator and judge, discarding all question-answer pairs with fewer than two source documents attributed by both LLMs, ensuring true multihop characteristic for the whole dataset.

3.2. Cited text span extraction

3.3. Span evaluation

4. Evaluation

4.1. Dataset

4.2. Cited text span extraction

4.3. Span evaluation

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A. Appendix

```
You will be be provided 5 scientific paper text, which share same topic that they are talking about. They will be provided in the following format:
[
  {
    "ArXiv": string,
    "text": string,
  }
]

You should do the following:
Step 1: Read the given scientific paper texts and extract a list of 10 topics where the papers overlap, contract or complement each other. Focus on important concepts or entities within the papers. Avoid using generic or broad words.
Step 2: Use the Topics from Step 1 to generate 1 scientific question-answer pair, with the following requirements.
Requirements:
-question should be based on the information provided in multiple of the papers
-try generating a questions, that requires multiple papers to answer
-question must be context independently answerable, so no reference to a specific paper or entities that you can only understand with the specific paper. The question should have the same character, as if you would ask an scientific expert in their field something, without having a specific paper in mind. Do not refer to external sources like figures or tables.
-question cannot contain explicit references to the papers or its content such as "in this paper", "the proposed methods" or similar
-prioritize a question that requires synthesizing, resolving, applying, or evaluating information across papers
-question should be a complex scientific question that needs in depth knowledge in that area, avoid just asking a simple or definitional question
-try answering the question as specific as possible
-add the arxiv ids of the papers, that contain the relevant information for answering the question
-only add the arxiv id of the paper if it really did contribute significantly to the answer of the question

provide your answer, stricly following this json format:
{
    "topic overlaps" : [
    <TOPIC OVERLAP 1>,
    <TOPIC OVERLAP 2>,
    ...
    ],
    "question": "<YOUR_QUESTION>",
    "answer": "<YOUR ANSWER>",
    "usageJudgementGeneratorLLM":[<ARXIV_ID_1>, <ARXIV_ID_2>, ...]
```

```
}
Do not deviate from this schema. Dont add the keywords you generated. Do not add any preciding information like ‘‘‘json. Only Answer with the valid json
Paper Texts: {paperTexts}
```

Listing A.1: Question–Answer generation prompt

```
You will be provided a scientific question, an answer, and a scientific paper text.
Your task is to evaluate whether the specific information in the scientific paper text is necessary to answer the question.

Step 1: Read the question and the synthesized answer. Identify the core scientific claims being made.
Step 2: Scan the scientific paper text.
Step 3: Determine if the paper provides explicit evidence, data, or mechanisms that directly support the answer. Do not accept mere keyword overlap.

Output your final verdict inside <verdict> tags. The verdict must be exactly "true" or "false".

Output Example:
    <Your thinking process described in step 1-3 here>
    <verdict>true</verdict>

question: {question}
answer: {answer}
scientific paper text: {scientificPaperText}
```

Listing A.2: paper usage judgement prompt